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Phase 2 Presentation of Major Project

Implementation Stage

AY : 2025-26

"Knack: A Gamified Platform for Skill Development"

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Items 13 to 15 cover what was added in this stage: mock interviews, company data, the group-discussion room, voice input, saved personalisation and offline mode.

INTRODUCTION

- **Overview:** Knack is a web-based, gamified career simulation that prepares engineering students for campus placements: a group discussion, five interview rounds, joining, an internship, an offer and full-time work.
- **Current Scenario:** Placements are decided in a few conversations, yet students rarely get to rehearse a group discussion or an interview under realistic conditions.
- **Relevance:** Communication, situational judgement and knowing how a specific company hires are what decide a shortlist.
- **Status at this stage:** The platform is **fully implemented and running**. A student picks a company, takes a mock before every round, joins a discussion with five profiled panellists, speaks answers aloud, and keeps practising offline.

PROBLEM STATEMENT

The Core Problem: Placement-preparation tools are either static question banks that never adapt or chatbots that adapt to nothing, and both usually need a network.

Key Issues:

- **No Group-Discussion Practice:** A discussion needs five or six people willing to act as candidates, so students cannot rehearse one alone.
- **Generic Preparation:** Question lists say nothing about how a particular company hires or what it tests.
- **Assessment Before Teaching:** Mock tools score answers without first showing what a strong answer looks like.
- **No Memory of the Learner:** Every session starts from zero, so practice does not build on itself.
- **Network Dependence:** AI tools stop working on a weak hostel network or an exhausted free quota.

OBJECTIVES

- **Simulate the Full Placement Path:** Group discussion, five interview rounds, joining, internship, offer and work in one journey, each stage unlocked on competency and tied to real company profiles.
- **Teach Before Testing:** Adaptive, AI-generated scenarios and an MCQ mock interview before every real round, with every option graded and explained.
- **Realistic Group Discussions:** Five profiled characters who speak one at a time, with a live reading of the mood of the room.
- **Habit and Measurable Progress:** XP, levels, streaks, stars and a level map, with per-topic skill scores and a Professional Readiness Score.
- **Personalisation Saved in the Backend:** Stored profile, history and rolling statistics choose questions, difficulty and panel pressure, at a bounded prompt cost.
- **Natural and Always Available:** Voice input, human-sounding AI text, and graceful degradation to a full offline mode.

OBJECTIVES — EVIDENCE IN THE CODEBASE

Objective	Where it is implemented	Source file	Status
Simulate the full placement path	Six phases with score gates (discussion 6/10, each round 6/10, internship 5/10); 16 company profiles with headcount, turnover, hiring steps and simulated openings	`career.js`, `companies.js`	Met
Teach before testing	84-item MCQ bank on 49 topics; the real round is refused (HTTP 409) until its mock is done; options shuffled, answer key kept on the server	`mock/`, `routes/mock.js`	Met
Realistic group discussions	Five fixed panellists with profiles and SVG faces; one speaker at a time; mood read from the last six lines	`gdRoom.js`, `Character.jsx`	Met
Habit and measurable progress	XP, named levels, streaks, 24 badges, level map and stars; rolling skill scores and a readiness score	`gamification.js`, `badges.js`	Met
Personalisation saved in the backend	Rolling mean of the last five scores per topic with a trend; weighted question choice; panel pressure from attributes; 60-word learner summary	`personalise.js`, `context.js`	Met
Natural and always available	Mic in every answer box; humanising post-filter; offline switch with a local bank, scripted panel and rubric	`speech.js`, `humanise.js`, `settings.js`	Met
Improvement shown with students	Not attempted yet. A pilot study with before-and-after ratings is planned	—	Not met

Every row above can be opened in the repository during the viva. The last row is stated as outstanding rather than claimed.

LITERATURE SURVEY

Paper Title	Key Findings	Limitations
MACH: My Automated Conversation Coach (Hoque et al., 2013)	A virtual interviewer that reads speech and facial behaviour gave automated feedback that improved interview performance.	One-to-one interview only; needs camera and audio sensing; no company context.
The TARDIS Framework: Virtual Agents for Social Coaching in Job Interviews (Anderson et al., 2013)	Scenario-based virtual recruiters can coach the social skills used in job interviews.	Covers the interview alone; no group discussion or workplace phases.
Coaching and Performance in Situational Employment Interviews (Maurer et al., 1998)	Candidates who attended coaching performed better in situational interviews.	Coaching by people; not self-paced or available on demand.
Empirical Evidence on Computer Games and Serious Games (Connolly et al., 2012)	Games and serious games show positive effects on knowledge and engagement.	Evidence is uneven and general; not aimed at placement preparation.
Does Gamification Work? A Review of Empirical Studies (Hamari et al., 2014)	Gamification mostly has positive effects on engagement.	Effects depend on context and users; engagement alone can reward the wrong thing.
The Gamification of Learning: A Meta-Analysis (Sailer & Homner, 2020)	Small to moderate positive effects on cognitive, motivational and behavioural outcomes.	Results vary with design; mechanics must be tied to performance.

LITERATURE SURVEY (CONTINUED)

Paper Title	Key Findings	Limitations
Distributed Practice in Verbal Recall Tasks (Cepeda et al., 2006)	Spacing practice over time improves retention compared with massed practice.	Studied on verbal recall, not on judgement or interview questions.
Test-Enhanced Learning (Roediger & Karpicke, 2006)	Taking a test is itself a learning event and improves long-term retention.	Uses memory materials, not scored workplace scenarios.
Knowledge Tracing (Corbett & Anderson, 1994)	Estimates mastery of each skill from the sequence of right and wrong answers.	Needs a detailed skill model and is opaque to the learner.
A Trainable Spaced Repetition Model for Language Learning (Settles & Meeder, 2016)	Half-life regression predicts recall to schedule reviews at scale.	Needs large usage data and a cloud service.
Local-First Software (Kleppmann et al., 2019)	Software should keep working, and keep data on the user's device, without the network.	A design approach; not applied to learning or AI tools.
A Circumplex Model of Affect (Russell, 1980)	Emotions can be placed on two axes: pleasure and arousal.	A psychological model; gives no way to read a group's mood from text.

These studies shaped the mock-before-assessment rule, the question-selection weights, the offline mode and the room-mood reader.

RESEARCH GAP

- **Fragmented Practice:** Group discussion, interview and workplace practice are offered separately, if at all.
- **Generic Employers:** Preparation rarely depends on the company the student is aiming at.
- **Assessment Before Teaching:** Mock tools score without first explaining what a strong answer sounds like.
- **Opaque or Absent Personalisation:** Adaptation is either missing or hidden from the student.
- **Network Dependence:** Generation and speech services assume a connection and a quota.
- **Generic Generated Text:** Model output carries stock phrasing that students quickly learn to ignore. No reviewed work combines a profiled group discussion, employer-specific preparation, mock-first teaching and an offline mode.

EXISTING SYSTEM VS PROPOSED SYSTEM

Aspect	Existing Systems	Proposed System (Knack)
Practice scope	Interview questions only	Group discussion, five interview rounds, joining, internship, offer and work
Group discussion	Needs five or six real people	Five profiled panellists who speak one at a time, with the room's mood
Employer context	Generic, the same for every company	16 company profiles: turnover, headcount, hiring steps, open roles
Assessment	A score without explanation	MCQ mock first, with every option graded and explained
Personalisation	None, or level selection only	Profile, GitHub, resume and stored rolling statistics
Input	Typing only	Typing or speaking through the mic
Generated text	Stock AI phrasing	Voice brief in prompts and a humanising post-filter
Availability	Fails when the service is down	Full offline mode, clearly labelled

SYSTEM REQUIREMENTS

Hardware Requirements

Processor	Dual-core x86-64, 2.0 GHz or above
Memory	4 GB minimum, 8 GB recommended
Storage	500 MB including dependencies
Network	Not required; used only for optional AI and browser speech
Microphone	Optional, for spoken answers
Client device	Any modern browser, phone or laptop

Functional Requirements

- Account creation and secure login
- Company selection and company profiles
- Mock interview before every real round
- Group discussion with the room's mood
- Voice input in every answer box
- Progress, history and training profile

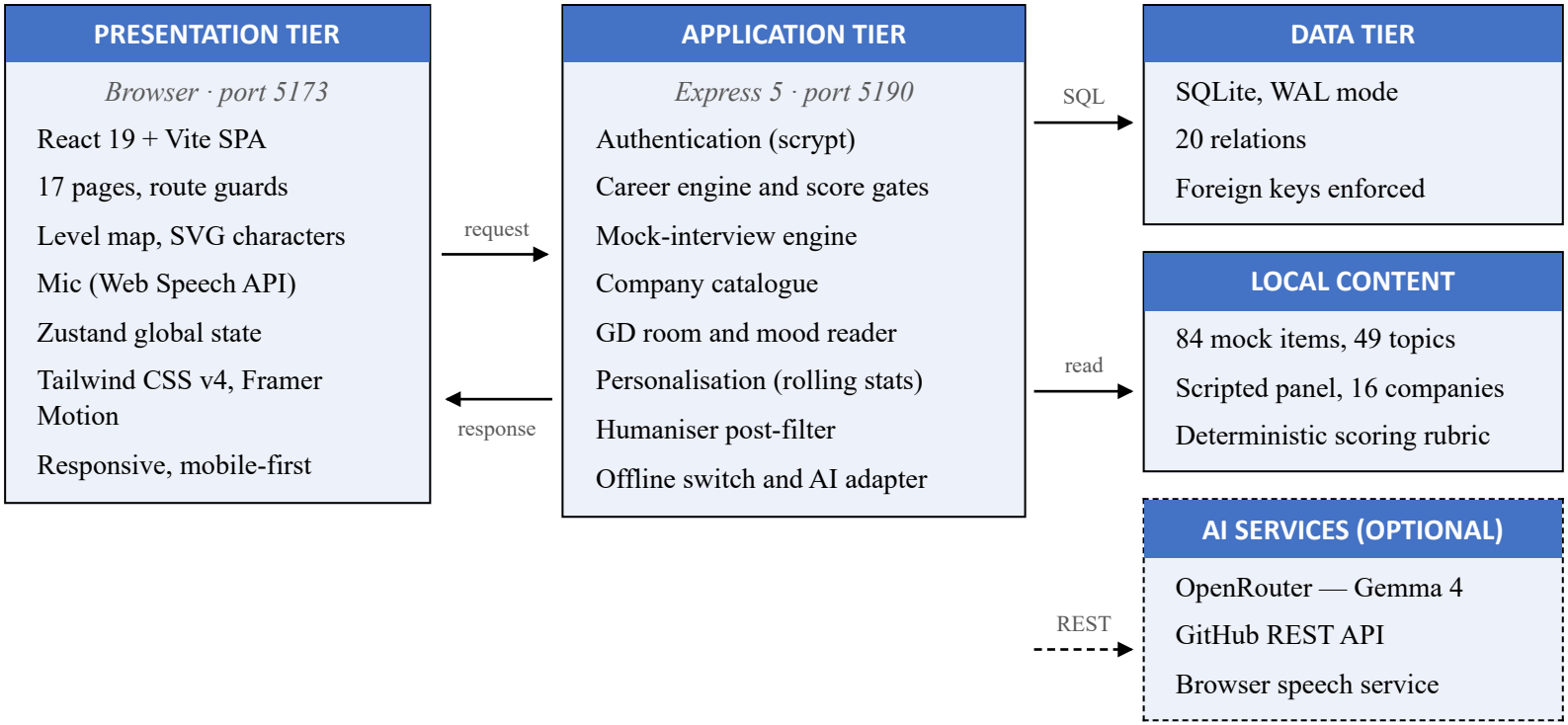
Software Requirements

OS	Windows 10/11, Linux or macOS
Runtime	Node.js v24 LTS
Database	SQLite (file-based, no server)
AI service	OpenRouter API key (optional)
Browser	Chrome or Edge, current version (needed for voice input)

Non-Functional Requirements

- **Offline-first** — every feature has a local fallback
- **Bounded cost** — prompt size independent of history
- **Security** — scrypt hashing, hashed session tokens
- **Isolation** — data scoped to the owning account
- **Speed** — about 1 ms median server response
- **Accessibility** — respects reduced-motion setting

METHODOLOGY / PROPOSED ARCHITECTURE



The application tier binds the loopback interface only. In offline mode, or when any AI call fails, the engines answer from local content, so every feature keeps working.

Figure 1. Three-tier system architecture

SYSTEM DESIGN AND FLOW DIAGRAM

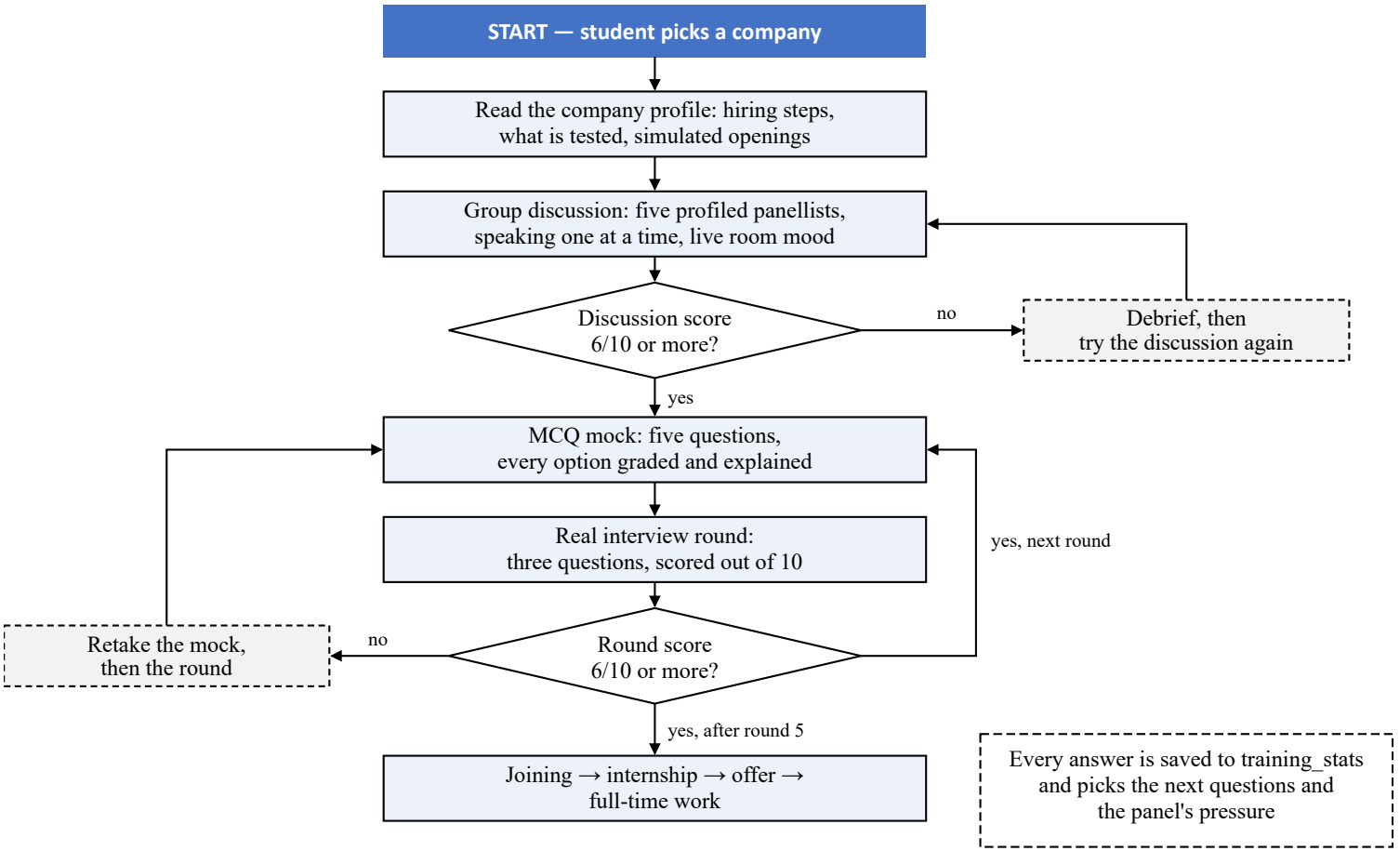
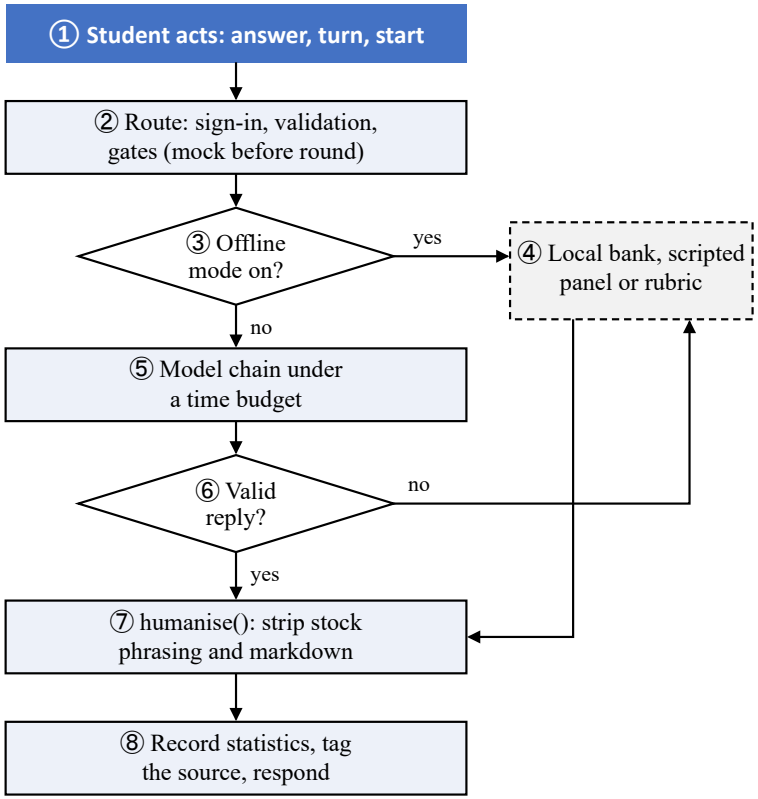


Figure 2. End-to-end flow. Progress is gated on scores, never on XP, and a mock always comes before a real round.

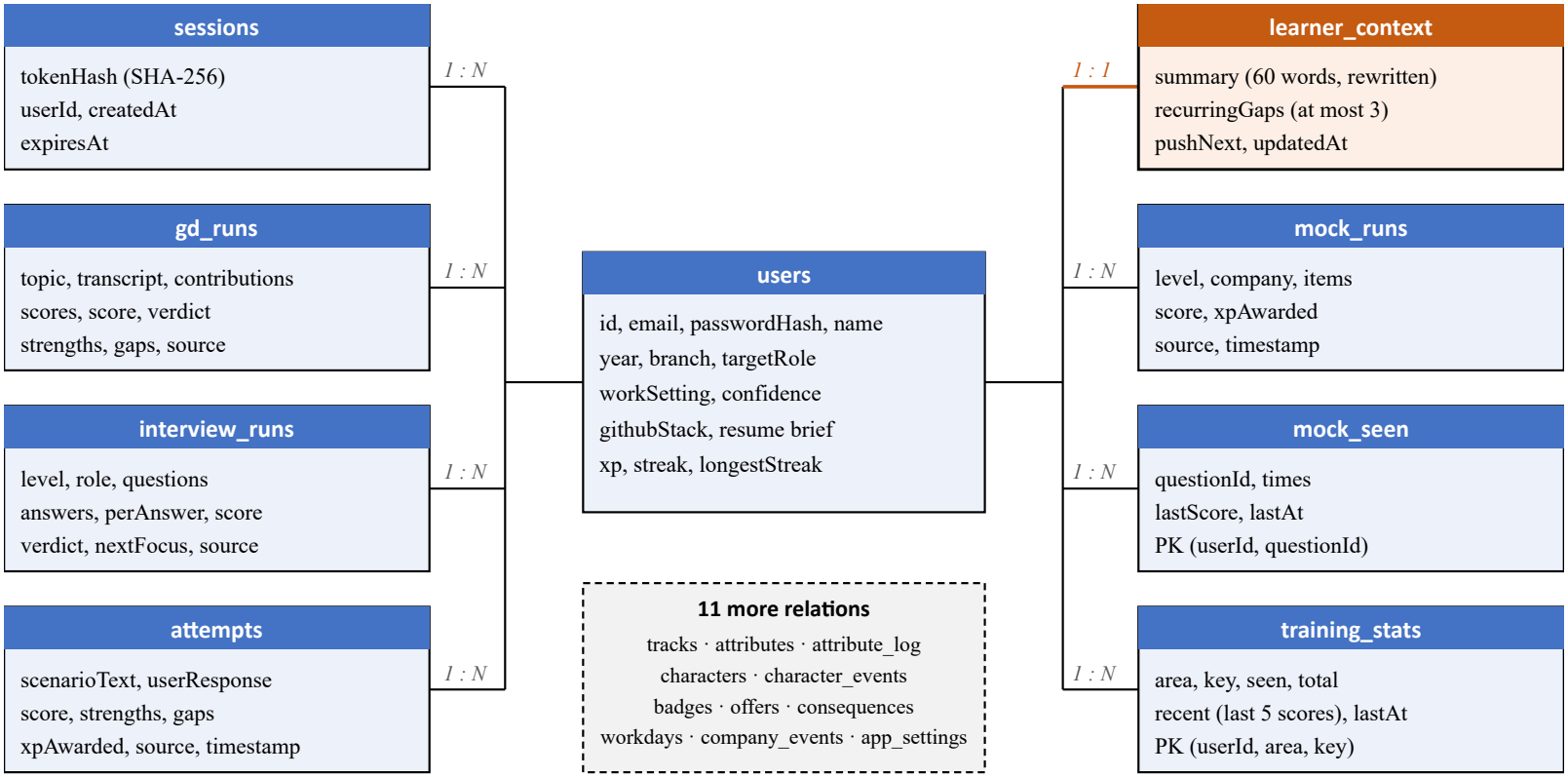
SYSTEM DESIGN — OFFLINE-FIRST REQUEST PATH



1. A session cookie is resolved to a user before any route runs.
2. Gates are checked on the server: a real round returns HTTP 409 until its mock is complete.
3. Offline mode is on if `KNACK_OFFLINE` is set (then it cannot be switched off in the app) or if the stored setting says so.
4. Local sources: the 84-item question bank, the scripted panel and a deterministic rubric.
5. One attempt per model down the chain, inside a time budget.
6. Any failure or malformed reply drops to the same local path.
7. Generated and scripted text both pass the same post-filter.
8. Statistics are written for personalisation, and anything not generated is labelled, for example “Scripted panel”.

Figure 3. Path of a request that may need a model. Both branches meet at the same post-filter and the same record step.

DATABASE DESIGN



mock_runs, mock_seen and training_stats are new in this stage: they hold the mock history and the statistics that personalise the next session.

Figure 4. Entity-relationship design (20 relations, WAL mode, foreign keys enforced)

MODULES AND THEIR FUNCTIONALITY

Module	Functionality	Key design decision
Authentication	Registration, login, sessions, protected routes	Only the hash of a session token is stored
Profile and Portfolio	Onboarding, GitHub reader, resume brief	Only bounded skill lists reach the prompt
Career Engine	Six phases, score gates, ten attributes	Progress is gated on scores, never on XP
Company Catalogue	16 profiles with hiring steps and openings	Openings are modelled and labelled simulated
Mock Interview	Five MCQs before each real round	The answer key never reaches the browser
Group Discussion	Five profiled panellists, one turn at a time	Silence is a real choice and is scored
Sentiment Analysis	Lexicon-based sentiment of each line: tension, warmth, energy	Rule-based, so it works with a scripted panel
Personalisation	Rolling per-topic statistics, question weights	Arithmetic on stored rows, shown in full
Humaniser	Strips stock AI phrasing and markdown	Runs on scripted and generated text alike
Voice Input	Mic button in every answer box	Browser speech; on-device when offline
Offline Switch	Local bank, scripted panel and rubric	Locked on when set by the environment

TECHNOLOGIES, TOOLS AND SOFTWARE USED

Layer	Technology	Purpose
Frontend	React 19, Vite 7, Tailwind CSS v4, Framer Motion, Zustand, React Router	Single-page interface, state, animation and routing
Backend	Node.js 24, Express 5, cookie-parser	REST endpoints, middleware and session handling
Security	scrypt from Node's built-in crypto module	Memory-hard password hashing
Database	SQLite via better-sqlite3, WAL mode	Local transactional storage, 20 relations
AI engine	Gemma 4 31B via OpenRouter, with configurable fallback models	Mock items, panel turns, grading and summaries (optional)
Speech	Browser Web Speech API	Spoken answers, on-device where available
Graphics	One parametric SVG character component	Faces and expressions for every person
Fonts	Fredoka and Nunito, bundled	No network needed to render the interface
External API	GitHub REST API v3	Reads public repository metadata for portfolio grounding
Tools	Git and GitHub, VS Code, Chrome DevTools	Version control, development and mobile testing

Complete source code: **github.com/ishaaqdev/Knack**

NEW FEATURES ADDED IN THIS STAGE

Feature	What was built	Source file
Voice input	Mic button in every answer box; on-device recognition requested in offline mode	`MicButton.jsx, speech.js`
Real company data	16 profiles with turnover, headcount, hiring steps, what is tested, tips and open roles	`companies.js`
MCQ mock first	Five questions before every real round, each option graded and explained	`mock/, routes/mock.js`
Panel speaks one by one	Five profiled panellists take turns; the student speaks to the room or to one person	`gdRoom.js, GroupDiscussion.jsx`
Mood of the room (sentiment analysis)	Lexicon-based sentiment analysis of the last six lines: tension, warmth and energy, and each person's mood and expression	`gdRoom.js, RoomMood.jsx`
Humanised AI text	Voice brief in every prompt and a post-filter on every string	`humanise.js, prompts.js`
Character profiles	Parametric SVG faces with a background, strengths, a tell and a way of speaking	`Character.jsx, PersonProfile.jsx`
Game-style design	Level map, star ratings and chunky buttons, on black with a single orange accent	`JourneyPath.jsx, Stars.jsx`
Saved and personalised	Rolling per-topic statistics choose questions, panel pressure and the next step	`personalise.js`
Offline mode	One switch, or `KNACK_OFFLINE`; local bank, scripted panel and rubric	`settings.js, fallback.js`

Every feature has a local fallback, so none of them depends on the internet.

IMPLEMENTATION DETAILS — MOCK INTERVIEW AND COMPANY DATA

- **Mock format:** five multiple-choice questions before each real round; four options scored 9, 6, 3 and 1, labelled Strong, Decent, Weak and Costly.
- **Teaching first:** every option is explained after the answer. Options are shuffled on every run, and scores are released from the server one question at a time.
- **Gate:** the real round returns HTTP 409 until its mock is complete; XP is paid only for the first mock of a round.
- **Question bank:** 84 hand-written items on 49 topics, with items for service, product, big-tech and start-up employers.
- **Company profiles:** 16 employers (13 real, 3 invented) with description, culture, headcount and turnover tagged by year, hiring steps, what is tested and tips.

$$O_r = \text{round}_2(\, N \cdot \lambda_t \cdot s_{t,r} \,)$$

Openings for role r: N headcount, λ_t yearly intake rate (services 7%, product 6%, big-tech 2.5%, start-ups 20%), $s_{t,r}$ share of the role. Labelled *simulated*; real vacancies are on the company's careers page.

IMPLEMENTATION DETAILS — GD AND SENTIMENT ANALYSIS

- **Panel:** Rohit (Dominant), Anjali (Knowledgeable), Vikram (Aggressive), Sneha (Diplomatic) and Imran (Quiet), each with a background, strengths, a tell and a way of speaking.
- **Turns:** they speak one at a time. The student speaks to the room or to one person, or stays quiet; never speaking caps every score at 25.
- **Pressure:** $v = (c + m) / 2$ from confidence and communication: gentle below 42, tough above 66, steady otherwise.
- **Sentiment analysis:** lexicon-based. Each of the last six lines is matched against word lists for pushback, agreement and bridging, and the counts give the room's mood. No AI is needed, so it works offline.
- **Faces:** each panellist's expression follows their last line: pushing back, agreeing, bridging or probing.

$$T = 15 + 16n_c + 4n_q - 9n_w - 8n_b$$

$$W = 35 + 14n_w + 14n_b - 9n_c$$

$$E = 25 + 8k + 5n_q + (\ell - 12)$$

Tension, Warmth, Energy, 0–100. n_c, n_w, n_b: pushback, agreement, bridging lines; n_q: questions; k: lines; ℓ: mean words per line.

Label	Condition	Label	Condition
Heated	$T \geq 66$	Lively	$E \geq 62$
Tense	$T \geq 45$, $E \geq 55$	Flat	$E < 38$
Warm	$W \geq 62$	Calm	otherwise

Loosely modelled on Russell's circumplex of affect.

IMPLEMENTATION DETAILS — PERSONALISATION

Every mock answer, discussion score and interview round writes one score from 0 to 100 to the **training_stats** table. Everything below is arithmetic on stored rows, so it works with no AI model.

- **Rolling score:** the mean of the last five scores per topic; the trend compares the newer half of that window with the older half once four exist.
- **Question selection:** unseen questions, weak topics and questions missed a while ago are weighted up; questions answered Strong or seen in the last six hours are weighted down.
- **Panel pressure and next step:** read from the same rows, and shown on the Profile page as “What Knack has learned about you”.
- **Learner summary:** a 60-word summary rewritten every three attempts when the AI is available; the prompt size stays constant.

$$w(q) = \max(\, 0.15,\ 1 + 2\cdot[\text{unseen}] + 1.5\cdot[\text{weak topic}] + \delta(q)\,)$$

δ: +1.5 if missed over 12 h ago · +0.3 if missed recently · −0.8 if last answered Strong · −0.6 if seen in the last 6 h

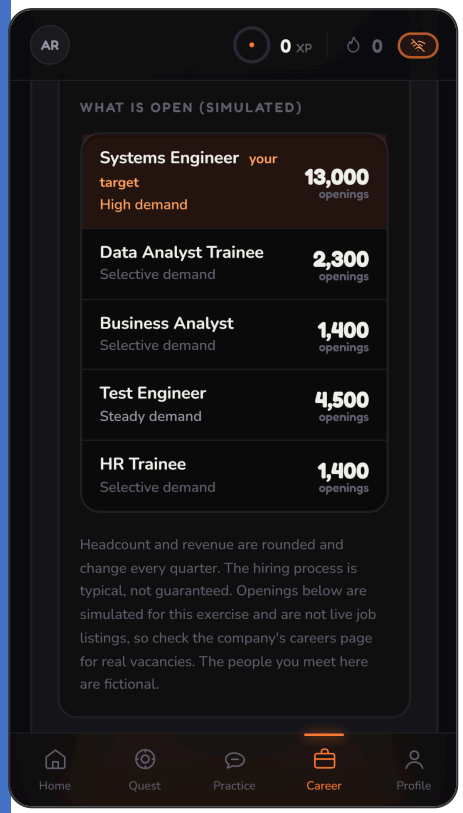
This is not RAG: there is no vector store or retrieval step. The final draw is random, so two students with the same history do not get the same five questions.

IMPLEMENTATION DETAILS — OFFLINE MODE, VOICE AND HUMANISED TEXT

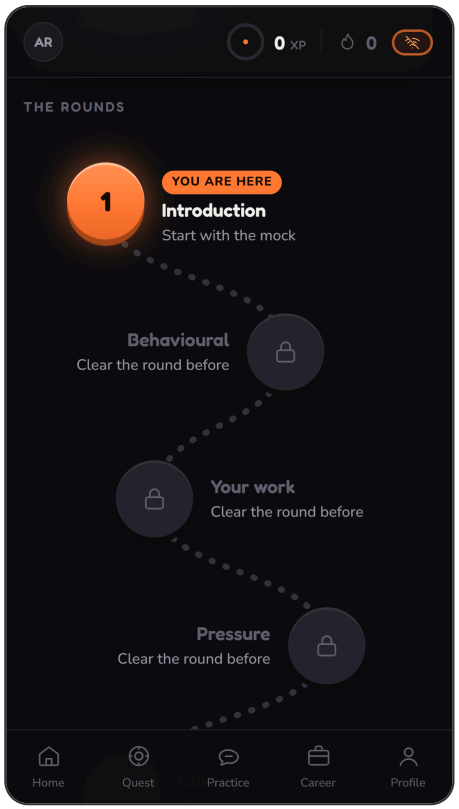
- **Offline switch:** `KNACK_OFFLINE=1` or a stored setting. The model client stops at once and each feature uses its local source: question bank, scripted panel, deterministic rubric.
- **Labelled honestly:** anything that was not generated is tagged in the database and labelled in the interface, for example “Scripted panel”.
- **Voice input:** the Web Speech API fills the answer box as the student speaks. In offline mode the mic asks for on-device recognition, and typing always works.
- **Clear errors:** a “network” error is explained as the browser's speech service being unreachable, not the student's internet.
- **Humaniser:** removes markdown and praise openers, swaps stock connectives (“Furthermore” → “Also,”, “utilize” → “use”) and rewrites dashes.

Before (model output)	**After the post-filter**
Great question! Furthermore, it is important to note that we should utilize a **balanced** approach — one that serves everyone.	Also, we should use a balanced approach, one that serves everyone.
Certainly! That is a valid point. Additionally, we can leverage existing infrastructure —	That is a valid point. Also, we can use existing infrastructure...

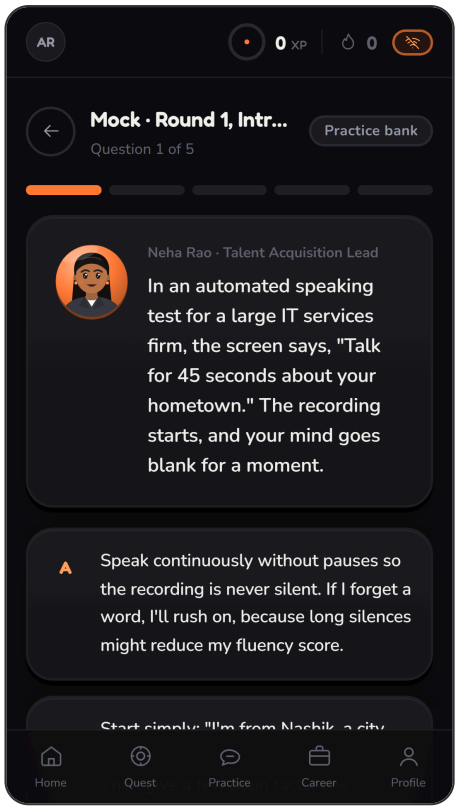
SCREENSHOTS OF IMPLEMENTED MODULES



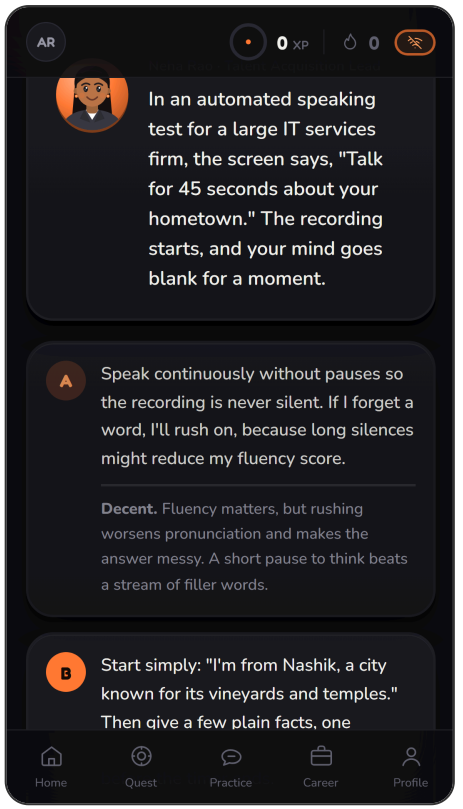
Company profile



Level map



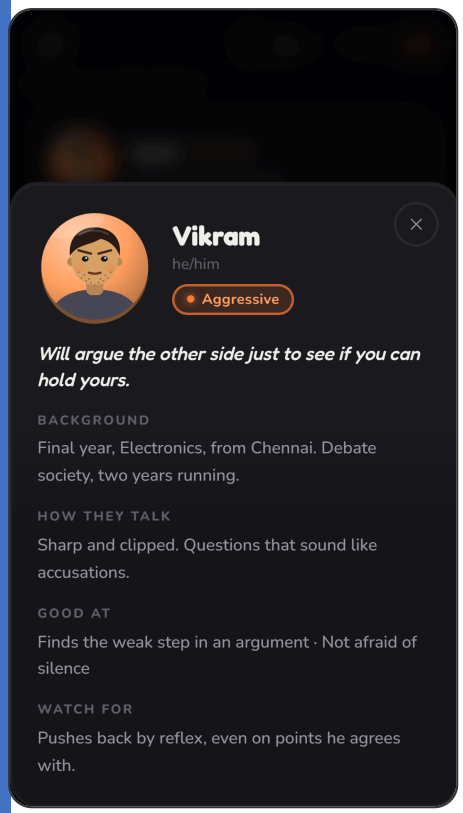
Mock question



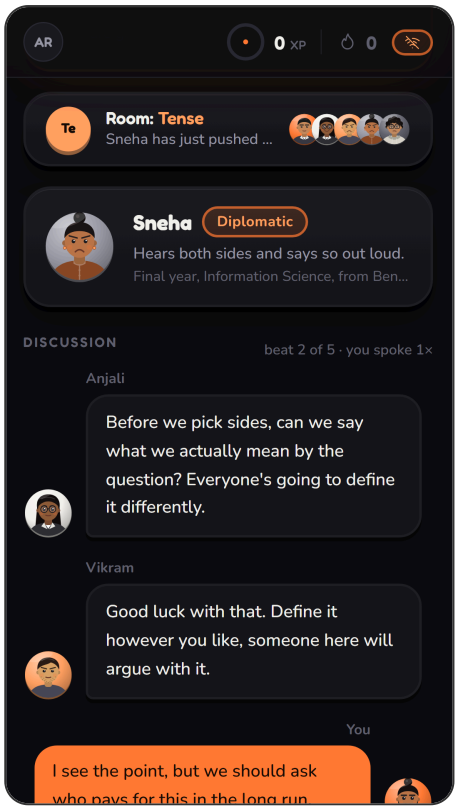
Every option explained

Real screens from the running app on a phone (9:16), in offline mode. Openings are labelled simulated.

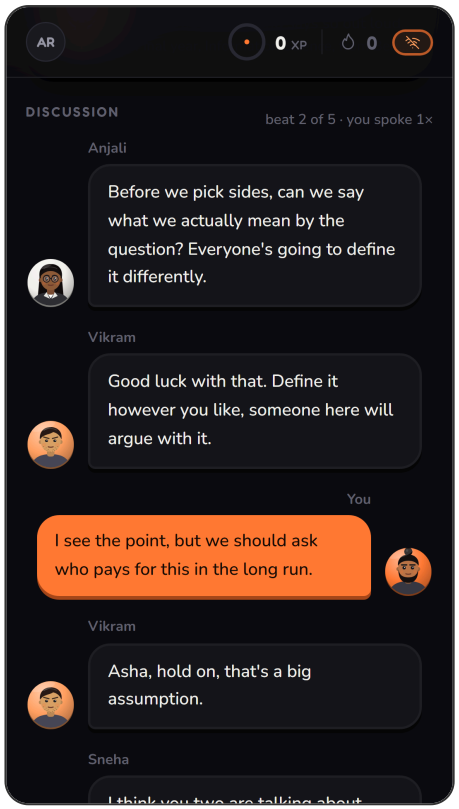
SCREENSHOTS OF IMPLEMENTED MODULES (CONTINUED)



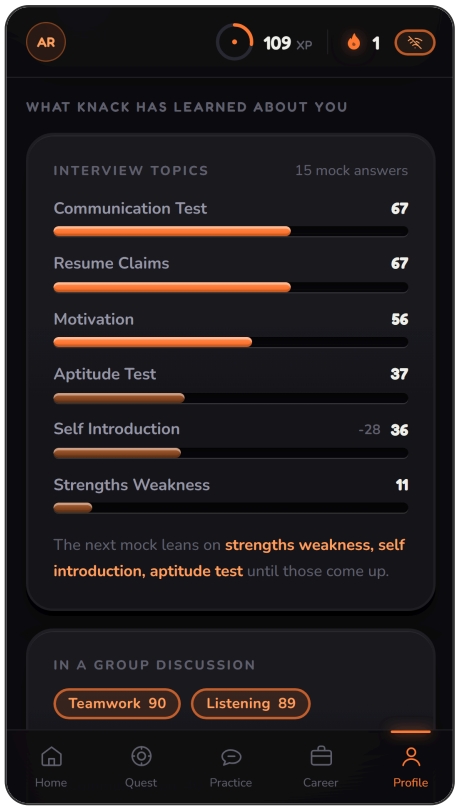
Panellist profile



Mood of the room



Group discussion



Training profile

The discussion here runs on the scripted panel; with an AI key the same screens show generated turns.

RESULTS / CURRENT PROGRESS

Property tested	Method	Observed result
End-to-end flows	Smoke test against a server in offline mode	**37 of 37** checks passed
Answer key hidden	Inspect the mock question payload	No scores or reasons before the student answers
Mock gate	Open a real round before and after its mock	Refused (HTTP 409) before; opens after
Response time	200 calls per endpoint, offline, on a laptop	Median 0.6–1.0 ms; p95 at most 1.7 ms
Question selection	2000 simulated learners against two baselines	**41% fewer** wasted repeats; **61% more** revisits of missed questions
Sentiment analysis	Four constructed transcripts	Heated, Warm, Flat and Flat, as intended
Humaniser	Four generated-style strings	Praise openers, markdown and stock connectives removed
Offline lock	Start with `KNACK_OFFLINE=1`, try to switch it off	Refused; the setting cannot be undone from the app

Stated honestly: the simulation uses an assumed learner model, and no study with students has been run yet. What is shown is that the system works as designed and works offline.

RESULTS — QUESTION-SELECTION SIMULATION

Policy	Coverage after 2 mocks	Coverage after 3 mocks	Wasted	Revisited	Mock 4 score (of 9)	Pool P(Strong)
Random	0.660	0.801	0.348	0.152	7.44	0.735
Unseen-first	0.833	1.000	0.267	0.145	7.37	0.751
Knack	0.736	0.928	**0.158**	**0.233**	7.31	0.752

- **Setup:** 2000 simulated learners, four mocks of each of the five rounds, one day apart. Knack's policy is the real selection code on a throwaway database.
- **Wasted** is the share of questions the learner had already answered Strong; **revisited** is the share they had last missed.
- **Finding:** Knack wastes 41% fewer questions and revisits 61% more missed ones than a no-repeat cycle, but simulated learning is the same (0.752 vs 0.751). The weighting buys focus, not faster learning.

CHALLENGES FACED AND SOLUTIONS ADOPTED

Challenge	Cause identified	Solution adopted
Free AI quota and outages	Every feature depended on the model	Offline switch; local bank, script and rubric for every call
Generic, formulaic AI text	Models default to stock phrasing	Voice brief in prompts and a rule-based post-filter
Answer key exposed to the browser	Scores sent along with the questions	Kept on the server and released one answer at a time
Mood note claimed attention that was not there	A generic “you” was read as addressing the student	Only a spoken name counts as addressing them
Mic said “no internet” on a working connection	The browser's speech service was unreachable and reported as “network”	The message now names the browser and the real cause
Discussion page went black for some names	The avatar picker used a signed shift on an unsigned hash	Unsigned shift, plus an error screen with a reload button
Fonts need the network	Fetched fonts fail offline	Fonts bundled with the client
Company figures change	Headcount and turnover vary by year	Rounded and tagged by year; openings labelled simulated

LIMITATIONS AND ASSUMPTIONS

- **Company figures are rounded** values from public disclosures and should be checked against the companies' own reports before they are quoted.
- **Openings are modelled.** The intake rates and role shares are assumptions, which is why every openings figure is labelled simulated.
- **The mock bank has 84 items**, so a student who retakes a round many times will see repeats.
- **The sentiment analysis is lexicon-based.** It misses irony and negation: “I don't disagree” still counts as pushback.
- **Voice input depends on the browser's speech service.** Firefox has none and Brave blocks it; typing always works.
- **No study with students has been run yet.** The claim is that the system works as designed and offline, not that it improves placement outcomes.

FUTURE ENHANCEMENTS

- **Pilot study with students.** Rate group discussions and interviews before and after, by raters who do not know the condition, and measure usability.
- **Spoken replies.** The interviewer and the panel speak back; dictation exists, speech output does not yet.
- **A larger mock bank and company catalogue**, with more employer-specific questions.
- **AI-based sentiment analysis** for the room mood that handles negation and irony, with the lexicon method kept as the offline fallback.
- **Richer learner models** behind the same question-selection interface.
- **Several companies at once and a native mobile client.** The interface is already responsive, so this is incremental rather than architectural.

CONCLUSION

- **Objectives met.** A working career simulation takes a student from a group discussion through five interview rounds to full-time work, tied to real company profiles.
- **Teaching comes first.** Every real round is preceded by a mock in which every option is explained.
- **The room feels real.** Five profiled panellists speak one at a time beside a live reading of the room's mood.
- **Personalisation is stored and visible.** Rolling statistics in the database choose questions and pressure, and the Profile page shows them.
- **It works anywhere.** Voice input, humanised text and a full offline mode; 37 of 37 checks passed offline with responses of about 1 ms.
- **Limitations are acknowledged.** No student study yet, and in simulation the selection rule buys focus rather than faster learning.

Complete implementation and documentation: **github.com/ishaaqdev/Knack**

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